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Online Shoping System

Class Identification and Abstraction.

Step-1 → Name of All 10 Classes.

(1)- Customer (2) Product (3) ShoppingCart
(4) Order (5) Payment (6) Delivery (7) Seller
(8) Review (9) Discount (10) Notification.

Step-2 → Attributes and methods For Three Important classes.

*1) Customer.

Attributes	methods.
• customerID	• registerCustomer()
• name	• login()
• email	• updateProfile()
• password	• placeOrder()
• address	

2* Product.

Attributes	methods.
• productID	• displayProductDetails()
• name	• checkAvailability()
• price	• updatePrice()
• description	• updateStock()
• stockQuantity	

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3. Order.

Attributes	Methods.
• orderID	• CreateOrder()
• customerID	• calculateTotal()
• OrderDate	• confirmOrder()
• totalAmount	• CancelOrder()
• OrderStatus	• trackOrder() .

Step-3: Abstraction Decisions.

→ In the Online shopping system, the product class stores important details such as product name, price etc but ignores unnecessary physical detail such as the product's manufacturing process because they are not required to complete an online purchase.

→ The customer class stores the customer's ID, name, email, but ignores unrelated personal information such as hobbies and favorite places.

→ The Order class focusses on order's ID, products total amount, and order status. but Ignores the customer's activities outside the Shopping System.

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Step-4 → Identify one invented Software Object.

∴ Invented Object :- OrderValidator.

* The ~~Order Validator~~ OrderValidator is a software object that does not exist as a physical object in the real world. It is created in the Online Shopping System to check whether an order can be placed Successfully.

OrderValidator.

Attributes	Methods.
• customerStatus	• ValidateCustomer ()
• productStock	• checkProductStock ()
• paymentStatus	• validatePayment ()
	• approveOrder ()

Example:-

when a Customer clicks the "place Order" button, the "orderValidator checks whether the customer is logged in, the selected products are available, and the payment information is valid. if all conditions are satisfied, the system allows the order to be confirmed.

